Blockchain & Distributed Ledger Observatory

Blockchain & Web3: time to build

2022-2023 Research Report

January 2023

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What happened in 2022 in the Blockchain and Distributed Ledger market?

What are the most recent technologic innovations in Blockchain and Distributed Ledger?

What are the main areas of application of Blockchain? What are the main international projects?

How much is the Blockchain and Distributed Ledger market worth in Italy?

2022 has been a year of extremes for Blockchain. On one hand there has been the dramatic burst of some financial bubbles and a prolonged depreciation of all crypto-assets – known as cryptowinter – on the other hand consolidation of the development work started years ago was completed, proof of unprecedented technical verve and engineering pragmatism.

Resulting from the combination of very different elements, this situation is often over-simplified in the representation made by general media, who lack appropriate vocabulary to distinguish fundamentally heterogeneous technologies, attitudes, and perspectives. Thus, while there is the all-inclusive "crypto-" prefix often used for every aspect of the phenomenon, encompassing the multitude of different circumstances and applications in one would be like mixing apples and pears.

Among the remarkable technological advancements in 2022 that testify to an evolution towards maturity, what undoubtedly stands out is the changes to Ethereum's consensus mechanism, that after years of development and caution transitions to the Proof of Stake responding to the omnipresent objection to Proof of Work 's energy consumption, thus proving the sensitivity of a significant part of this industry towards sustainability issues. Significant progress has also been made in terms of performance and costs and other traditionally problematic issues owning to what is known as layer 2 approaches, that following the experimental status became a tangible reality.

Amid the most interesting application areas, after the speculative bubble that brought NFTs to mainstream media, "classic" platforms like Instagram have introduced tokens into their own worlds, exposing them to a very large audience. In a "core" application field such as disintermediated value transfer, crypto-assets have emerged as an interesting alternative to Ukraine's traditional instruments for economic support, enabling to collect tens of millions of dollars from small donors around the world also as a result of a record NFT sale ($6.5M for a token of the Ukrainian flag). Another important indication of the increase of such instruments is the highest value transacted through stablecoins, that reached a record 7 trillion $.

There have also been spectacular failures, such as the collapse of FTX, one of the major centralized exchanges widely employed by retail users, that resulted insolvent in November. The FTX case, although it could be "simply" ascribed to typical embezzlement, has contributed to making the reputation of crypto-assets more problematic in the eyes of mainstream audience. But it is important to observe how the possibility for the FTX administrator to embezzle funds is due precisely to an excess of that inscrutable centralization to which blockchains offer a radical alternative.

The positive lesson that can be drawn from this suggests making better use of the possibility of implementing mechanisms that are more transparent than those "traditionally" imaginable in centralized systems.
But decentralization is not a cure-all remedy, as demonstrated by the Terra-Luna case that was based on a mechanism that provided dubious returns linked to a stablecoin – that by its nature should have been non-volatile – in a simply unsustainable system but – in full

DeFi style – in plain sight. Although decentralization did not prevent the market collapse, it did create an opportunity for unprecedented public debate on the underlying mechanisms, which allowed careful investors to escape in time.

These scandals, that have also affected small investors, have prompted calls for regulatory intervention to ensure clarity, security, and confidence, as well as to promote positive development. Such arbitration, however, could inhibit innovation or cause to waste great potential if conducted without any knowledge of the peculiarities of these technologies (for example, programmability, with which it is possible to create transparency, and fundamental and unprecedented guarantees).

So how to look at the future of these technologies, from the recesses of this cryptowinter made of highs and lows? Even if "financial" crises cause bad publicity, they create the conditions that promote development and production, moving away from highly speculative dynamics. The time to build is now!

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Blockchain & Web3: time to build

RESEARCH

January 2023

1. Preface

For the Blockchain industry 2022 has been especially marked by ups and downs. Global macroeconomic turmoil and some major events on the cryptocurrency market have resized the euphoria of 2021 and led to a new cryptowinter[1], following the one of 2018.

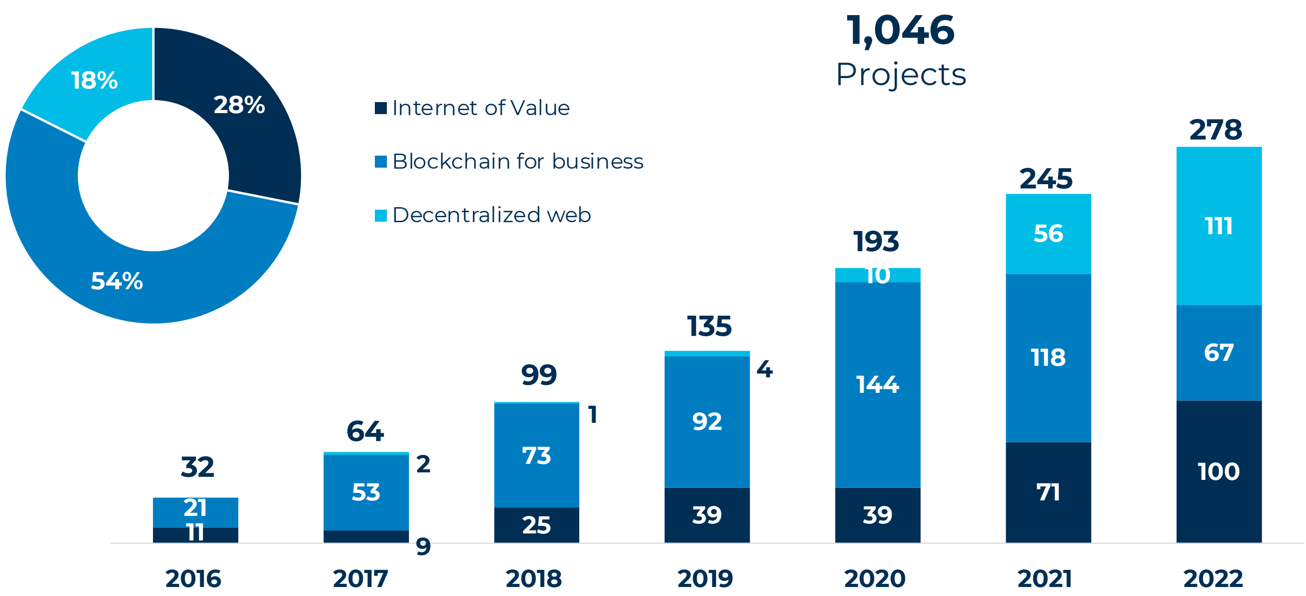

Fig 1. International Blockchain & Distributed Ledger projects divided into three categories per year (Base: 1,046 projects) / Source Digital Innovation Observatories – Politecnico di Milano (www.osservatori.net)

1. **Note**| Cryptowinter is an enthusiasm cool-down period connected to a decrease in the market value of cryptocurrencies and tokens.

However, considering the market performance, it doesn't seem the adoption of blockchain projects has slowed down. The Observatory identified 278 Blockchain projects of companies and PS bodies in 2022 worldwide, up 13% compared to the 245 registered in 2021.
A total of 2,033 global projects were surveyed between 2016 and 2022, of which 1,046 are established projects (either trials or already operative). Today, therefore, companies are leveraging the moment to focus on experimenting with web3 projects, far from the media hype and the most speculative dynamics. The Blockchain world is entering a new phase: the hype is over, the "time to build" is now.

Three types of technology applications can be identified: Internet of Value, ***Blockchain for business, and Decentralized web***[2].

2. Note| Internet of Value. Applications focused on value exchange (generally cryptocurrencies, stablecoins, and virtual coins promoted by central banks – CBDCs). These applications introduce a new way of managing valuable assets without intermediaries / Blockchain for business. Projects in which traditional business processes are replicated using Blockchain technologies. In this category projects generally pursue the following objectives: data verifiability, coordination and implementation of reliable processes/ Decentralized web. Decentralized applications (DApps), including collectibles. Blockchain becomes an enabling infrastructure for creating and developing innovative business solutions.

2. Internet of Value: on the winding way to legitimacy

Internet of Value (IoV) applications deal with value exchange as regards the use of cryptocurrencies, stablecoins, and CBDCs. Despite the cryptowinter, IoV projects continued to grow and today account for 28% of total surveyed applications.

These solutions are in the maturing process and are aiming for legitimacy, leveraging the strong interest of institutional and traditional players and the regulatory course of recent years. However, in 2022 two important events occurred that have put the confidence of companies and consumers to the test, with the risk of slowing down or interrupting the course taken: the collapse of the Terra-Luna ecosystem[3], and the downfall of the FTX exchange[4]. Clearly not attributable to a technology malfunction but resulting from mismanagement and no supervision over the Terra-Luna and FTX reserves, these two events contributed to the collapse of the cryptocurrency market, with a negative impact on sector companies.

However these facts have not stopped companies that are continuing to explore this industry. More than half (59) of the world's top 100 banks[5] have launched at least one project related to the use of stablecoins, CBDCs, or cryptocurrency storage and investment services. **Merchants and traditional companies that choose to accept payments in cryptocurrencies are also increasing**, even though they often rely on third-party services that convert the payments into fiat currency at the time of payment. Google, McDonald's,

3. Note| The collapse of the Terra-Luna ecosystem happened in May 2022, after the UST stablecoin lost its dollar-peg stability (pegged to the US dollar value).

4. Note| In just a few days FTX went from being worth $32 billion to bankruptcy. The downfall of the exchange also impacted on other players in the industry such as Block-Fi, that also went bankrupt a few weeks later.

5. Note| Total Assets Under Management.

Checkout.com, Emirates Airline, Gucci are just some of the players that have opened up to cryptocurrency payments in the last year.

Digital currencies are also at the base of Government and central banks CBDC testing. The Digital Euro is halfway through the investigation phase started in October 2021. Currently, the use cases identified by the ECB as urgent, concern payments in shops (physical and online), and P2P transfers[6]. To develop the first Digital Euro user interface prototypes, the ECB has selected 5 companies[7], including the Italian Nexi.

The road to creating a digital form of currency that is legally recognized and usable on Blockchain platforms (cryptocurrencies, stablecoins or CBDCs), has only just started and there is still uncertainty about which instruments will be validated first, and therefore authorized for use by web3 businesses.

In this scenario, regulation will play a crucial role. Europe has officially approved the Markets in Crypto-Assets regulation (MiCAr), defining a framework to regulate crypto activity markets and introducing rules for crypto-asset related issuers and service providers. Italy too is paying attention to this issue, by establishing a special section of the OAM (Agency for Credit Agents and Brokers) register aiming to list all providers of digital currency related services operating in Italy. More recently, the Italian government has included cryptocurrencies for the first time in the 2023 Finance Act, defining them as financial assets and introducing some aspects related to the current tax system.

6. Note| https://www.ecb.europa.eu/paym/digital_euro/investigation/governance/shared/files/ecb.degov220929.en.pdf

7. Note| Nexi will develop front-end prototypes for digital euro physical store payments initiated by the beneficiary. The other selected companies are Amazon for e-commerce payments, CaixaBank for P2P online payments, Worldline for P2P offline payments, EPI for payments in physical stores initiated by the payer.

3. Blockchain for business: struggles of building, maintaining, and expanding ecosystems

Despite the steep decline compared to 2021 (-43%) *Blockchain for business* projects still represent the majority of cases surveyed at international level (568 projects, 54% of total). The companies that develop these projects aim to enhance the efficiency of existing processes using Blockchain technologies to simplify data access and sharing (in 56% of cases), to ensure the transparency and immutability of information (38%), or to achieve reliable processes through smart contracts (6%).

The difficulties of these projects continued in 2022. The B3i[8] project defaulted and was shut down in July, as a result of failing to collect funding, followed in November by the end of Trade Lens – the document traceability in intermodal transport ecosystem project led by Maersk and IBM.

The closing down of such activities is not a symptom of the failure of a technology, but of the complexity of such large ecosystem projects, which often encounter difficulties on some key aspects, such as the creation of the ecosystem, economic sustainability, the development of evolutionary perspectives for the business ecosystem and the limited benefits brought by solutions based on private and permissioned distributed Ledgers.

8. Note| Project founded in 2016 by a consortium of insurance companies with the aim of improving the efficiency and reliability of reinsurance processes between multiple players through distributed ledger technologies.

4. Decentralized web: time to build value

2022 registered a strong growth of *Decentralized web* projects (184 cases surveyed, 111 in 2022, +98% compared to 2021). These projects are closest to the web3 paradigm and include the development of decentralized applications (DApps) and the many activities related to NFT.

The NFT trend originated with the creation of digital collectibles. A condition that since 2021 has generated a whirlwind of attention and attracted large capital investments, leading some digital artwork to be sold at record figures, thus generating further media coverage. Such bustle has attracted many traditional companies who, for the first time, approached Blockchain through activities related to NFT and digital collectibles. For many organizations NFT projects for the issuance of digital assets have provided a gateway to web3, also as a result of their ease of implementation. **More and more often companies try to build around NFT business strategies that include access to exclusive services or experiences in the metaverse**. In particular, many players in the fashion and luxury industry (such as Dolce and Gabbana, Nike, and Gucci) have launched projects to issue assets, both "phygital" and purely digital, building new business strategies around NFT.

Decentralized applications also continued to evolve. In particular, **DAOs and distributed governance systems on Blockchain have caught the attention of some traditional**

companies that are looking into how to leverage them to get users more involved. To date, however, these participatory and decentralized decision-making models are still used almost exclusively by the more mature DApps, such as DeFi apps. It is precisely the decentralized finance ecosystem that is confirmed to be among the most advanced permissionless Blockchain DApps having maintained its considerable importance despite the market difficulties.

In this area it will be important to understand how the legislator's position will evolve, who at the moment has not yet issued laws or bills to regulate this area but observes its developments with perplexity.

5. Blockchain platforms: the infrastructure for the development of web3

2022 was characterized by a remarkable development of Blockchain platforms particularly focused on increasing scalability and reducing energy consumption. Ethereum has completed the transition of its consent mechanism from Proof of Work to Proof of Stake, reducing the energy consumption of the validation process by over 99%. In this way, the problem of the high environmental impact of the Blockchain, that often conflicted with the requirements for the development of company use cases, has been superseded. Meanwhile, other platforms are consolidating their ecosystems; this is the case of BNB Chain, which has become the main Blockchain in terms of number of DApps and active users, outweighing the ecosystem of Ethereum applications. Platforms such as BNB Chain have created an authentic operating standard consisting of creating platforms with extremely low fees but less secure and decentralized and promoting the development of DApps often copied from other platforms by providing developers with substantial funding.

6. The Italian market: growing business and consumer adoption

After the wait that characterized the past years, in 2022 there has been a sharp increase in corporate Blockchain projects in the Italian market. Investments by Italian companies grew considerably (+50%), reaching 42 million euro in 2022. The highest investments are in the financial and insurance sector (which accounts for 36% of total), the major development of 2022 is the growth of the retail and fashion sector, with 23% of the value invested as a result of the many NFT and digital collectible related projects. In third and fourth place are the automotive industry and the Public Sector, which respectively collect 10% and 7% of the total market. In line with international trends, Italy registers a growth of NFT issuance related activities that have attained great attention and driven many companies that had not previously invested in this technology, closer to the Blockchain.

The increased spread in Italy of Blockchain and its applications does not involve just companies In fact, more than 7 million Italians[9] have already purchased cryptocurrencies or tokens, plus another 7 million Italians claim to be interested in doing so in the future. The most adopted methods to come into possession of these resources are cryptocurrency exchanges (40%), followed by cryptocurrency ATMs (19%), and wallet services that enable direct purchase (18%). 52% of Italians chose to acquire cryptocurrencies and tokens indirectly through traditional financial trading services and the applications of their

9. Note| Research conducted on a representative sample of the Italian population with internet access between 18 and 75 years of age.

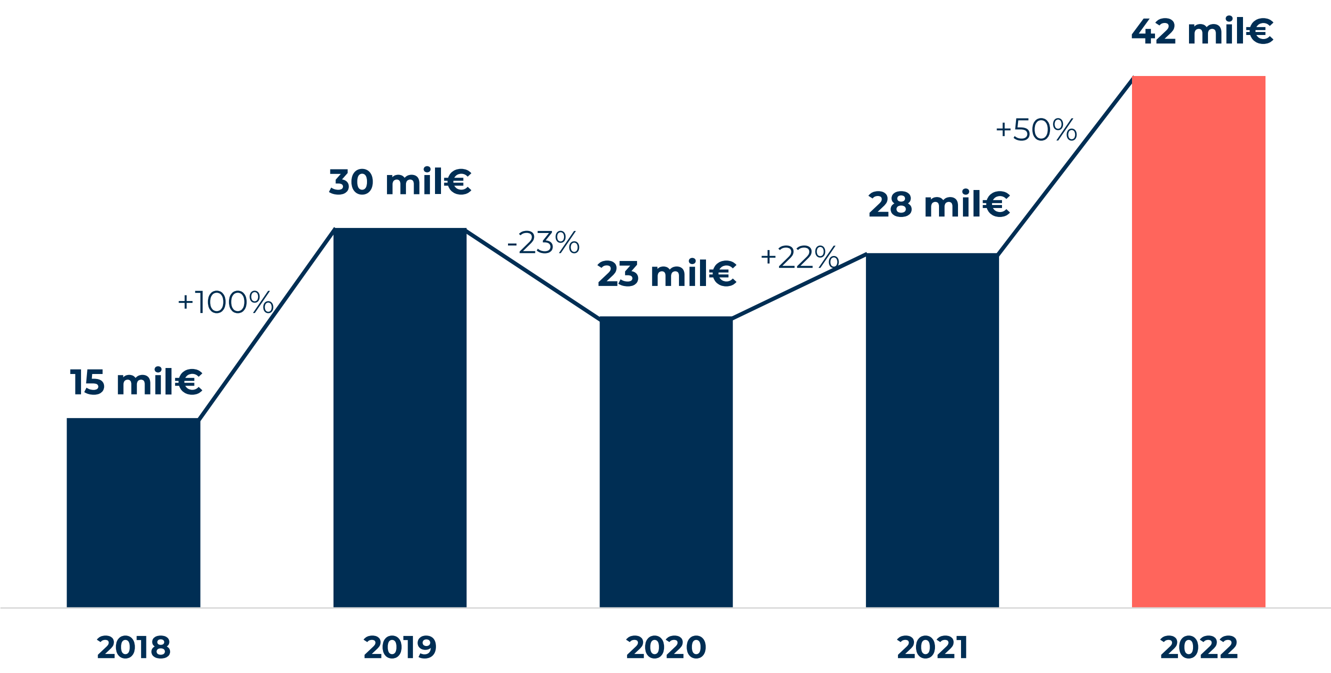

Fig 2. Investments in Blockchain and Distributed Ledger projects in Italy (estimate by the Blockchain & Distributed Ledger Observatory) / Source Digital Innovation Observatories – Politecnico di Milano (www.osservatori.net)

own banks. Cryptocurrency exchanges are also the storage method preferred by more than half of Italians, who, however, also widely use non-Custodial wallets. The 3 most used exchanges by Italians are Coinbase, Crypto.com, and Binance. If we consider exclusively NFT ownership the numbers are lower, with 9% of Italians who at the moment claim they have bought them and 14% who intend to buy them in the future. The NFTs most purchased by Italians are those related to digital artwork, avatars, and collectibles.

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January 2023

2022 Research conducted by the Blockchain & Distributed Ledger Observatory focused on studying Blockchain technologies and their potential business implications. In keeping with the previous edition, the 2022 edition focused on 15 lines of research divided in 3 areas:

- Business research (Projects and application areas in Italy and worldwide, Italian market, Startups and innovative trends, Internet of Value, Opportunities offered by tokenization, Blockchain ecosystems, DApps and DeFi, Web3, Decentralized Governance, Decentralized Metaverse)
- Technology research (DAO, DApps, and programmability, Data privacy, Performance, Scalability and energy consumption, Interoperability, Platform comparison, Convergence with other technologies).
- Legal research (Regulation and legislation)

Following is a description of the main methodologies.

Projects in the world [Business area]. A non-exhaustive survey was conducted to identify major announcements and implementation projects (including proof of concept, pilot or operational projects) with international relevance (local announcements were therefore excluded) concerning the use by national and international companies and PSs of Blockchain or Distributed Ledger solutions. 2,033 cases were identified worldwide between 2016 and 2022 (of which 406 in 2022).

Italian market [Business area]. An estimate of the investments and the number of projects conducted by companies in Italy has been made through 15 interviews with the leading Blockchain solution providers. A non-exhaustive survey was also conducted through an online survey the main implementation projects in Italy concerning the use of Blockchain or Distributed Ledger solutions by Italian companies and PS bodies, with over 252 Italian cases identified between 2016 and 2022 (value not methodologically comparable to the international survey). A non-exhaustive census of 129 Italian startups was then performed (value not methodologically comparable to the international startup survey). An online survey was also conducted in collaboration with DOXA, on a national sample representative of the 18-75 year old Italian internet population, divided by age, gender, geographical area, size of the municipality of residence, education, and occupation.

Innovative startups and trends [Business area]. A non-exhaustive census of Blockchain and Distributed Ledger startups was conducted to identify the innovation guidelines in effect. 1,683 international startups founded between June 2017 and June 2022 were analyzed and the analysis was performed on a sample of 1,410 startups that received funding for at least 1 million euro between June 2020 and June 2022. Over 15 interviews were conducted with international and national startups. Preliminary results were discussed with experts during a dedicated workshop.

Internet of Value [Business area]. A non-exhaustive census was conducted at international level on 139 use cases of cryptocurrencies and stablecoins by companies and public sector bodies between 2013 and 2022 (included in the global census). An analysis was performed on the top 100 international banks by managed asset value according to S&P Global as of April 11/2022, and 91 cryptocurrency, stablecoin, and CBDCs adoption projects have been identified. An analysis of the main scientific articles, international reports and informative articles, on a total of over 35 texts was conducted. A non-exhaustive survey was conducted on the progress status of adoption project plans of a Central Bank Digital Currency by the world's leading central banks, in particular, 92 initiatives that evaluate the adoption of Blockchain and Distributed Ledger technologies were analyzed. Preliminary results were discussed with experts during a dedicated workshop.

Opportunities provided by tokenization [Business area]. A non-exhaustive census was conducted on 259 cases at international level of the use of tokens developed with Blockchain technology between 2019 and 2022 (included in the global census). An analysis of the main scientific articles, international reports and informative articles was also performed on a total of over 30 texts. Preliminary results were discussed with experts during a dedicated workshop.

Web3, DApp, and DeFi [Business area]. An analysis was conducted on main applications by number of users. The census involved the top 100 DApps by Active Monthly

Users (AMU); DApp mining took place in April 2022 using the DappRadar site as main source. The data obtained was then enriched through secondary sources such as the application websites and blogs, the news published, and the market related figures according to the CoinGecko website. In order to better understand the composability, tokenomics, and governance trends within the DApp ecosystem and the web3 world, an analysis of secondary sources was conducted, that led to examining over 70 documents including papers, specialist reports, and informative articles. Preliminary results were discussed with experts during a dedicated workshop.

Platform analysis [Technology area]. The evolution of the main platforms was monitored, and various approaches to overcoming the classic limits of the Blockchain and Distributed Ledger platforms were analyzed, focusing in particular on scalability, interoperability, and data privacy. The analysis was performed by studying over 30 articles and papers from secondary sources and through the development of proof of concept and prototype protocols. A DAO demo was held during a dedicated workshop.

Analysis of the regulatory environment [Legal area]. The research activity focused on the main regulatory trends involving the Blockchain area overall. The chosen study approach provides a simple and pragmatic way of updating the main legal issues that emerged in recent years. The analysis conducted in 2022 has progressively given up on the "country-centric" perspective and is more transversal as this was considered more adequate to seize the recent trends and look at the bigger picture, orienting learners to draw their own reasoned conclusions. 2022 Research involved both the regulatory laws (or proposals) and the related scientific texts complementing or interpreting them. The most renowned newspapers were also taken into account to obtain an "A-technical" look and a general understanding of regulatory trends. Preliminary results were discussed with experts during a dedicated workshop.

Established in 2018, the mission of the **Blockchain & Distributed Ledger Observatory** is to generate and share knowledge on Blockchain and Distributed Ledger related topics and contribute to the development of the Italian market, creating debate, and meeting, and comparison opportunities for the main players that are active in this field.
2021 has been a year of maturation for the Blockchain and Distributed Ledger world. Technologies proceeded in their evolution and transformation journey while events, such as the booming of NFT and the growth of DApps and DeFi, captured the interest of institutions, companies, and media across the world.
Today, these technologies are no longer merely tools to improve existing processes but enable the creation of innovative and unprecedented organizational models. Together with other technologies Blockchain could be at the base of the "next web revolution", in other words Web3, a sort of decentralized web that could become the natural evolution of the current, centralized and dominated by big techs web.
Due to the collaboration between the Management, Economics and Industrial Engineering Department and the Department of Electronics, Information, and Bioengineering, the Observatory analyzes these topics from both a business and technical perspective.
Furthermore, in 2021 – from the Blockchain & Distributed Ledger Observatory – the Blockchain Innovation & Solutions HUB was founded, representing an impartial player that aims to promote, facilitate, and enable Blockchain projects involving different players through the creation of a precompetitive development environment where companies can conduct applied research activities.

2022 Research on Blockchain & Distributed Ledger

The 2022 edition, in keeping with the previous edition, will further develop the following areas of research:

6 lines of business research:

- Application environments in Italy and the world: analyze international projects as well as how Blockchain and Distributed Ledger technologies are used in projects by companies around the world.
- Italian market: estimate investments and number of

projects implemented by companies in Italy; determine the level of knowledge and adoption of Blockchain and Distributed Ledger technologies in Italian companies.
- Startups and innovative trends: analyze startups across the world and the spread of collected funding; description of the Italian startup ecosystem as regards Blockchain.
- Opportunities offered by crypto assets: analyze token application opportunities to identify the potential of digital assets.
- Blockchain Ecosystems: analyze main Italian and international ecosystem projects based on Blockchain and Distributed Ledger technologies. Identify major benefit categories that characterize the use of the technologies, and current chief limitations to the development of such projects.

6 lines of tech research:
- DApp and programmability: analyze leading decentralized applications on the market by number of active users, study their tokenomics and adopted business models.
- Data privacy: identify modes enabling to share data and information in Blockchain and Distributed Ledger platforms.
- Performance and scalability: identify the differences and ways of filling the gap between current performances of existing platforms and requirements of various applications.
- Interoperability: analyze main approaches to interoperability between different platforms, and potential benefits.
- Comparison of platforms: continuously monitor major and most interesting platforms and compare most relevant features/characteristics.
- Convergence with other techs: monitor interaction of Blockchain and Distributed Ledger technologies with other technologies (i.e. AI, IoT)

2 lines of legal research:
- Compliance and regulation: analysis of regulatory developments in countries already included in the Research Framework in addition to new ones.
- Analysis of some vertical matters on new topics that will be defined by the Research team in collaboration with research supporters.

4 vertical research streams:

- Identity: analyze main Blockchain digital identity solutions and the opportunities provided. Study the potential for businesses and public sector bodies of the Self Sovereign Identity model.
- Supply Chain: analyze application of Blockchain and Distributed Ledger solutions within the production chains of various sectors.
- Web3 & DApp: study new decentralized web models and the business opportunities provided by decentralized applications.
- Tokenization: analyze primary tokenization applications and solutions, and the possibilities offered by digital assets.
- DeFi: analyze main decentralized finance sector DApps available on the market, by total value locked.

In order to encourage an ongoing dialogue with institutions, associations, and research centers, an **Advisory Board** will be defined with the aim of directing research on the most valuable trends for companies and institutions. The Advisory Board will consist of a small group of organization representatives.

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INFRATEL ITALIA INVITALIA
INPS
Ministero dello sviluppo economico
OECD Better Policies for Better Lives
Regione Lombardia

Digital Innovation Observatories

In 1999, the Digital Innovation Observatories of the School of Management of Politecnico di Milano were established to raise cultural awareness in all the main areas of digital innovation. Today, the Observatories provide an expert point of reference for digital innovation, integrating Research work, Communication and Continuous Update. According to the vision driving the work of the Observatories, digital innovation is a key element of a country's development.
The mission is to create and spread knowledge on the opportunities and the impact of digital technology on companies, public authorities and citizens through interpretative models based on sound empirical evidence combined with spaces for independent, ongoing, and precompetitive discussions aimed at bringing together the demand-and offer-sides of Digital Innovation.

Activities are conducted by a team of nearly 100 professors, researchers and analysts in about 50 Observatories addressing key Digital Innovation topics in Companies (including SMBs) and Public Authorities

Digital Transformation:
Design Thinking for Business | Digital Agenda | Digital Innovation in SMEs | Digital Transformation Academy | Hi-tech Startup | Platform Thinking HUB | Smart Working | Startup Intelligence

Digital Solutions:
5G & Beyond | Artificial Intelligence | Augmented Reality & Metaverse Observatory | B2c eCommerce | Big Data & Business Analytics | Blockchain & Web3 | Cloud Transformation | Cybersecurity & Data Protection | Data Center | Digital B2b | Innovative Payments | Internet of Things | Mobile B2c Strategy | Multichannel | Omnichannel Customer Experience | Quantum Technologies | Space Economy

Verticals:
B2b Customer Experience | Business Travel | Cloud in the PS | Connected Car & Mobility | Contract Logistics "Gino Marchet" | Digital Content | Digital Export | Digital Healthcare | Digital Identity | Digital Innovation in Cultural Heritage and Activities | Digital Innovation in Retail | Drones and Advanced Air Mobility | EdTech | eGovernment | Fintech & Insurtech | Food Sustainability | HR Innovation Practice | Industry 4.0 Transition | International Observatory on Electronic Invoicing | Internet Media | Life Science Innovation | Professionals and Digital Innovation | Smart AgriFood | Smart City | Smart Working in the PS | Supply Chain Finance | Tech Company | Travel Innovation

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SCHOOL OF MANAGEMENT DEL POLITECNICO DI MILANO

The School of Management at the Politecnico di Milano opened in 2003, gathering various research, training, and consultation activities in the field of economics, management, and industrial engineering, which the Politecnico carries forward through its various internal structures and consortia. The School of Management holds the "Triple crown", the three most prestigious accreditations for business schools on a worldwide level: EQUIS, received in 2007, AMBA (Association of MBAs) in 2013, and AACSB (Advance Collegiate Schools of Business), obtained in 2021. In 2017, it became the first Italian business school to receive recognition for the quality of its courses provided through digital learning in its Executive MBAs using EOCCS (EFMD Online Course Certification System). Included since 2009 in the Financial Times list of the Best Business Schools in Europe, today it is listed with the full-time MBA, Master of Science in Management Engineering and Online MBA. In particular, the International Flex EMBA ranks 6th in 2022 among the best masters degrees in the world in the Financial Times Online. MBA Ranking. The School is also present in the QS World University Rankings and Bloomberg Businessweek Ranking. It is a member of PRME (Principles for Responsible Management Education), Cladea (Latin American Council of Management Schools), and QTEM (Quantitative Techniques for Economics & Management Masters Network). The School encompasses the Department of Management, Economics and Industrial Engineering at the Politecnico di Milano and the POLIMI Graduate School of Management, which is focused in particular on executive training and its masters programmes. Within the field of digital innovation, the School of Management's work consists of: Observatories for Digital Innovation, where research falls under the umbrella of the Department of Management Engineering, Executive education and Masters programmes, delivered by POLIMI Graduate School of Management.

Layout:
Emanuela Micello, Danilo Galasso, Miguel Luis Armenio and *Stefano Erba*

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